

Report Title: Cybersecurity Lab Build Report – Task 2

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Program: Cybersecurity Analyst Internship

Company: Maincrafts Technology

1. Executive Summary

****Objective:**** To design and deploy a secure, isolated lab environment for ethical hacking and security testing.

****Scope:**** Setting up an attacker machine (Kali Linux), a vulnerable target application, and segmented networking.

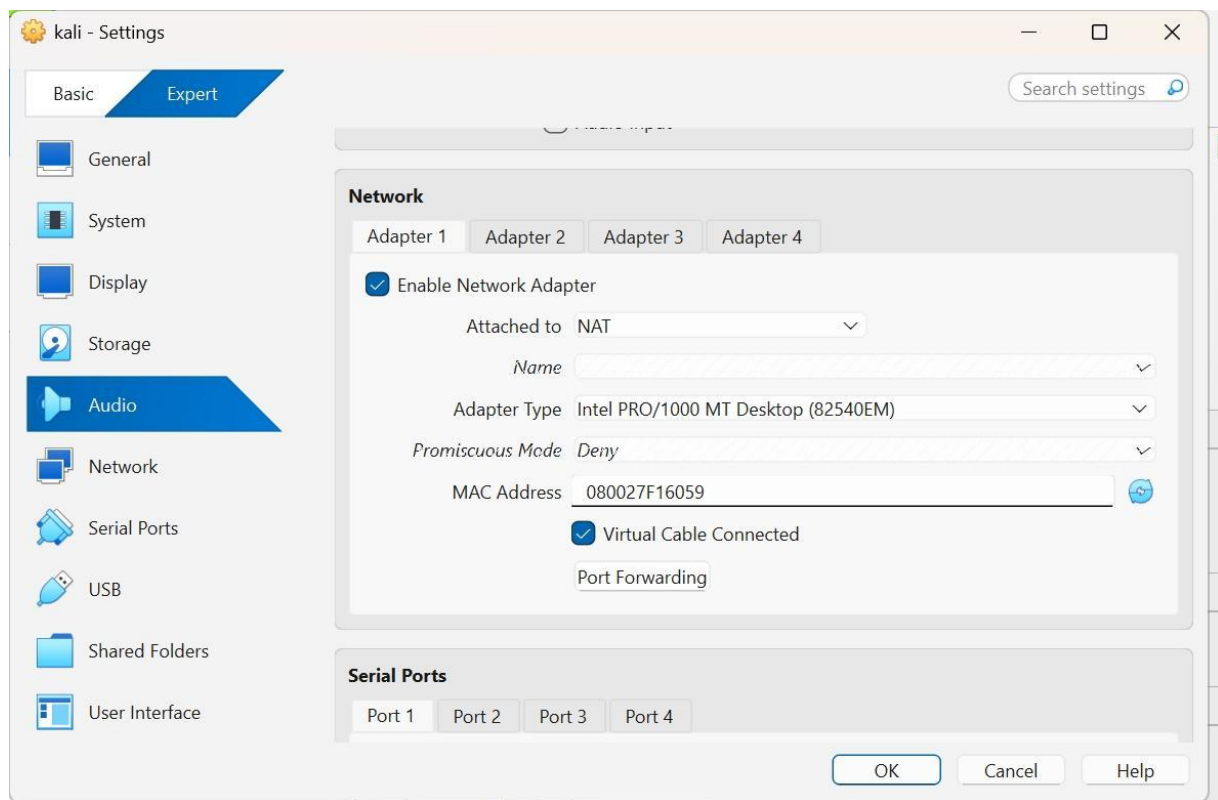
2. Lab Architecture

****Virtualization Platform:**** [Mention if you used VirtualBox or VMware].

****Network Segmentation:****

****NAT:**** Used for external internet access and system updates.

****Host-Only:**** Used for isolated communication between the Attacker VM and the Target VM.



3. Attacker Machine (Kali Linux)

* **Configuration:** Allocated resources (e.g., 2 CPUs, 4GB RAM).

* **Initialization:** Successful system update and creation of workspace directories (`~/lab/scans`, `~/lab/notes`, etc.).

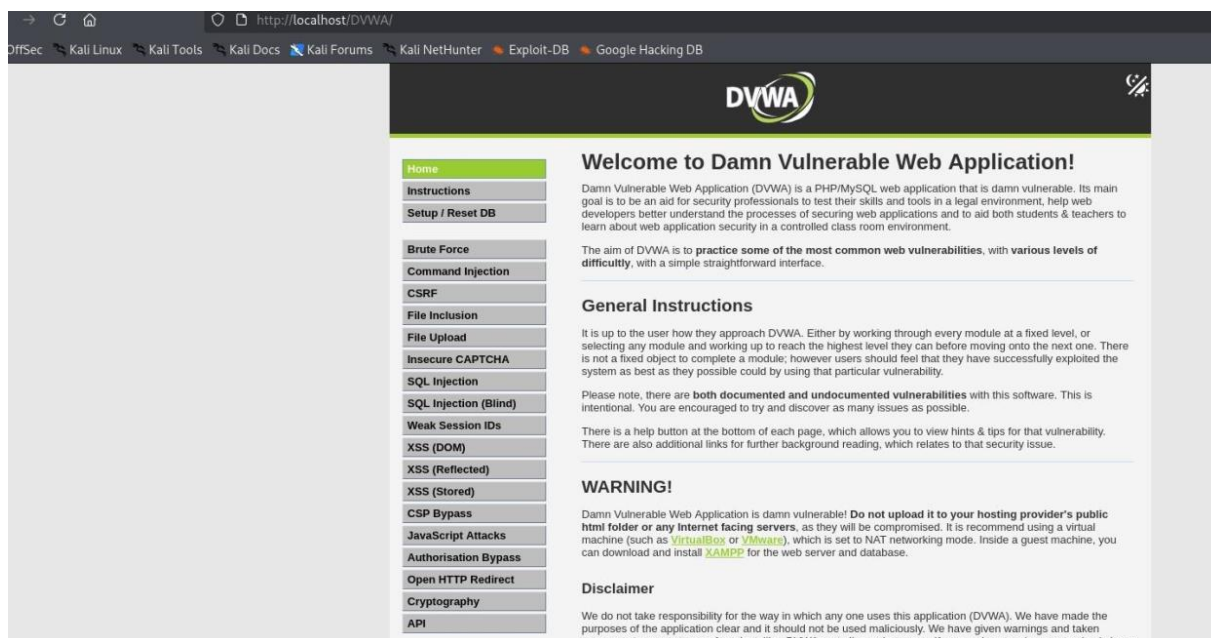
```
Session Actions Edit View Help
hero@hero: ~
(hero@hero)~-[~]
$ ls
Desktop Documents Downloads Music Pictures Public sample.txt Templates Videos

(hero@hero)~-[~]
$ sudo apt update && upgrade
[sudo] password for hero:
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.3 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
61% [3 Contents-amd64 23.1 MB/53.4 MB 43%] 1,919 kB/s 16s
```

4. Target Machine (Vulnerable App)

****Method Used:**** [State whether you used OWASP Juice Shop or DVWA].

****Deployment Status:**** Confirmation that the service is running and accessible



5. Connectivity & Validation

****Nmap Scan:**** Verification that the attacker machine successfully detects the target machine on the internal network.

****Ping Test:**** Confirmation of successful communication between VMs.

```
$ nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-01 10:04 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
3306/tcp  open  mysql
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

```
$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.637 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.054 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.138 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.053 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.112 ms
```

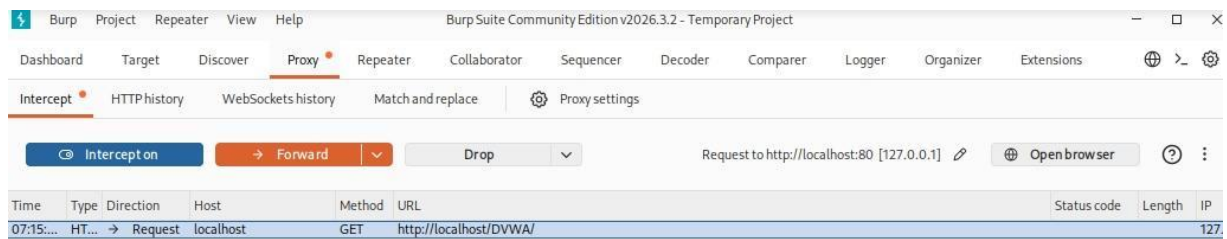
6. Security Tool Integration

****Burp Suite:**** Evidence of traffic interception.

****Wireshark:**** Evidence of packet capture within the lab environment.

The screenshot displays a Kali Linux desktop environment. On the left, the Wireshark network protocol analyzer is open, showing a packet capture of HTTP traffic. The packet list pane shows several HTTP GET requests to a DVWA instance. The packet details pane for the selected packet (No. 121) shows the raw packet data in hexadecimal and ASCII, along with a line-based text representation of the HTML content.

On the right, a web browser window is open, displaying the DVWA (Damn Vulnerable Web Application) interface. The browser's address bar shows the URL `http://localhost/DVWA/vulnerabilities/brute/`. The DVWA page has a sidebar with navigation links and a main content area titled "Vulnerability: Brute Force". The "Login" form is visible, with fields for "Username" (containing "admin") and "Password" (containing "*****"). The "Login" button is present. Below the login form, there is a "More Information" section with links to external resources.



7. Conclusion

* Summary of skills validated: Virtualization, network segmentation, and hands-on vulnerability research.

* The environment is now ready for future penetration testing and incident response tasks.